Linear Accelerator Shielding Specifications

<u>Accelerator/Room:</u> Varian True Beam/Linac vault

References to barrier numbers in this report supersede all references in previous reports.

Design Criteria

The following shielding calculations and specifications are derived based on the recommendations of the National Council on Radiation Protection and Measurements (NCRP) as contained in NCRP Report Number 151, *Structural Shielding Design and Evaluation for Megavoltage X- and Gamma-Ray Radiotherapy Facilities*, and *Shielding Techniques for Radiation Oncology Facilities*, by Patton McGinley.

No responsibility is assumed or implied for the adequacy of the enclosed specifications for workloads and/or conditions other than those stated herein.

* *The difference between this version of the shielding report and the previous version (#3b) is the change in barrier 5. Following a change in the occupancy factor, additional lead was added to this barrier, i.e., the lead thickness increased from 2" to 2 - 1/8 " . (Refer to images below Table 1)*

Introduction

<u>Manufacturer:</u> VM Systems, Inc. <u>Model</u>: True Beam

<u>Megavoltage of X-ray beam:</u> 15 MV, 10 MV, 10 FFF, 6 MV, and 6 FFF

<u>Workload Assumptions:</u>

Primary Barrier Workload: 150 Gy/week at 15 MV, 125 Gy/week at 10 MV, 125 Gy/wk at 10 FFF, 200 Gy/wk at 6MV, and 200 Gy/week at 6FFF

Leakage Workload: 450 Gy/week at 15 MV, 375 Gy/week at 10 MV, 375 Gy/wk at 10 FFF, 600 Gy/wk at 6MV, and 600 Gy/week at 6FFF

IMRT factor: 3

Total Body Irradiation (TBI) was not considered in this shielding design.

X-ray head leakage: 0.1%   Neutron head leakage: 0.05%

Total number of patients per day: 45

Use Factors

Primary Barriers	Use Factor
Floor	0.25
Ceiling	0.3
90^0	0.25
270^0	0.25

Shielding Material Densities

Material	Density (lbs/ft^3)
Concrete	147
BPE	59.2
Earth	93.4
High-Density Concrete	300
Lead	709
Steel	490

The vault is on a concrete slab on grade. The drawings provided by the facility are assumed to be complete and accurate.

Assumptions:

1. A four-inch instrumentation conduit is recommended to be routed from the case work at the console down under the floor using smooth bends and up in the therapy room on the other side of the console barrier.
2. Utility access for HVAC is provided over the maze door.
3. Drawings provided by the architects were assumed to represent the radiation treatment facility accurately.
4. The primary barriers will be centered on the isocenter.
5. No change of the isocenter location or additional penetrations will be made in the shielding without review by a qualified radiological physicist.
6. A qualified radiological physicist must review the final drawings for the facility before construction begins.
7. Concrete barriers must be poured using a vibrator to prevent voids or air pockets. When possible, the concrete should be continuous to eliminate seams. Ordinary *dry concrete* must have a density of at least 147 pounds per cubic foot. No cold joints are permitted in the construction of the vault.
8. Concrete thickness is normally specified in 147 pounds per cubic foot. When heavy (or high density) concrete is specified, it is absolutely essential that the cured density meet or exceed the specified density. Samples of each mix should be obtained and analyzed to ensure that the specified density is maintained. Specified thickness whether concrete, lead, high density concrete or composite thereof must extend to the footing.

9. Generally, all accelerator facilities are provided with laser alignment units for the purpose of indicating the center of rotation for the accelerator. If recesses in the concrete are specified for the laser units, steel or lead plates must be provided to compensate for the absence of the concrete in the recess.
10. Conduits, water pipes, etc. must not penetrate the shielding walls above grade. These should be routed under (below grade) the walls or through the utility access provided over the maze door.
11. Composite barriers of concrete/steel or concrete/lead require precautions in the placement of the lead or steel inside the concrete barrier. Unless otherwise indicated, a minimum of 2.5 feet of concrete will be required on the outside of the treatment room in all locations where lead or steel is used, in order to attenuate neutrons produced in these materials by the high-energy photon beam.
12. Conduits must be provided for two closed circuit TV cameras. The normal location of these cameras is approximately 8 feet above the floor on the barrier wall at the end of the treatment couch and adjacent to the primary barrier (thick wall section) on the left side of the treatment unit.
13. Convenience outlets, switch boxes, etc. should be surface mounted on shielding walls behind the sheetrock and furring strips. All recessed utility boxes must be provided with appropriate shielding.
14. Refer to the accelerator manufacturer's specifications for details of base frame, conduits, water pipes, etc.
15. Dual (two) micro switch interlock switches must be provided to operate with the opening of the therapy room door.
16. **In this report, the isocenter of the TB linac is located 4 feet 3 inches above the finished floor, 10 feet 1 inch measured from the concrete interior side of primary barrier 1, and 9 feet 5 inches from the concrete interior side of secondary barrier 3. This presents a shift in the location of the isocenter compared to the low-energy linac. Exchanging a 6MV linac with a high energy linac (6MV, 10MV, and 15MV) would lead to increases in thickness of primary and secondary barriers, door, and neutron beam shielding.**

    Significantly based on workload and the occupancy factor of surrounding space, the <u>typical</u> thickness of the primary barrier in a high-energy linac ranges from 4.9' to 9.8'. That for secondary barrier is 3.3' to 7.4. The linac door thickness, from inside the vault to outside, should be 0.2" to 0.8" of lead + 0.8 to 1.6" BPE + 0.4" lead.

    In Barrier #4, the centering of the existing steel barrier to the new isocenter is unnecessary. The existing concrete thickness (3'), steel barrier (height x high x depth: 3'-6" x 8'-6" x 8") and the low occupancy factor of that sidewalk lead to adequate rad shielding below the dose limit.

<u>Shielding Specifications Summary</u> (rev 3)

Facility: PAC Health Medical Center Unit: Varian True Beam

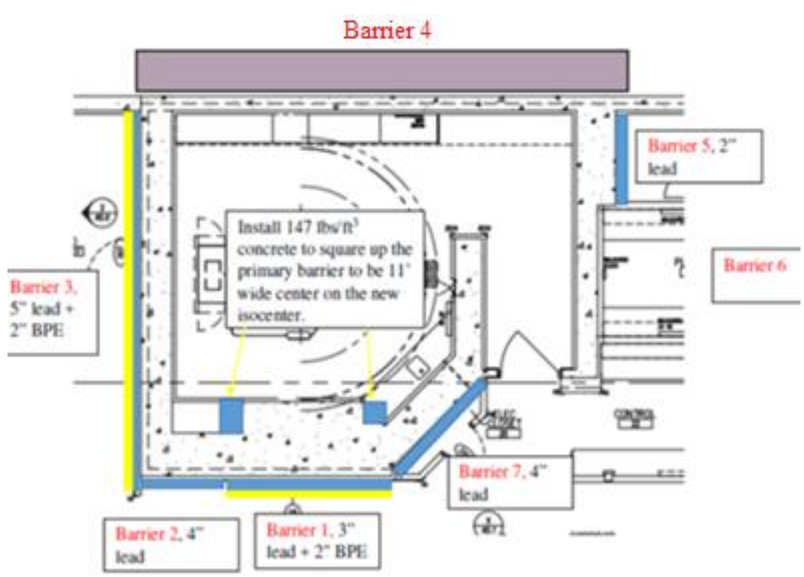
Table 1, below, summarizes the shielding requirements for each listed barrier and supersedes all individual barrier specifications. Any differences, whether larger or smaller, between barrier specifications and those presented in the table result from other considerations such as continuity of wall sections, predicted weekly exposure, and/or complex multiple alternate calculations. In the table, the column entitled "Barrier Thickness" lists the minimum amount of shielding required to meet regulatory requirements. These values are based on ALARA and/or cost considerations. The "concrete" below is *dry concrete* with a measured density of at least 147 lbs/ft^3.

Barrier[1]	Type[2]	Existing Barrier Width	Existing Barrier	Additional Shielding Required?	Comments
		Concrete @ 147 lbs/ft^3	Concrete @ 147 lbs/ft^3		
1	P	8'	64"	Yes	Install 3" lead + 2" BPE on the outside of the concrete vault. Extend from the floor to the top of the vault. Extend 11' wide centered on the isocenter. Inside the vault square up the primary barrier to be 11' centered on the new isocenter. Use concrete.
2	S	N/A	36"	Yes	Install 4" lead on the outside of the concrete vault. Extend from the floor to the roof of the vault. Extend the entire length of the barrier.
3	S	N/A	24"	Yes	Install 5" lead + 2" BPE on the outside of the concrete vault. Extend from the floor to the roof of the vault. Extend the entire length of the barrier. The existing brick vernier may be removed to accommodate the above.
4	P	N/A	24" + CAP flashing	No	See Figure 1.
5	S	N/A	36"	Yes	Install 2" – 1/8 " lead on the outside of the concrete vault. Extend from the floor to the top of the vault (see figure 2b). Extend the entire length of the barrier.
6	S	N/A	24" + 24"	No	

Barrier[1]	Type[2]	Existing Barrier Width	Existing Barrier	Additional Shielding Required?	Comments
		Concrete @ 147 lbs/ft^3	Concrete @ 147 lbs/ft^3		
7	S	N/A	36”	Yes	Install 4” lead on the outside of the concrete vault. Extend from the floor to the top of the vault. Extend the entire length of the barrier.
8 (Roof)	P	8’	44”	Yes	Install 26” concrete on top of the concrete vault centered on the isocenter 11’ wide. This concrete must extend the entire length of the vault from primary barrier to primary barrier. Install 20” concrete on bottom of the concrete roof on each side of the primary barrier so that the new primary barrier is at least 11’ wide, centered on the isocenter.
9 (Roof)	S	N/A	24”	Yes	Install 20” concrete on the entire concrete vault except for the primary barrier.

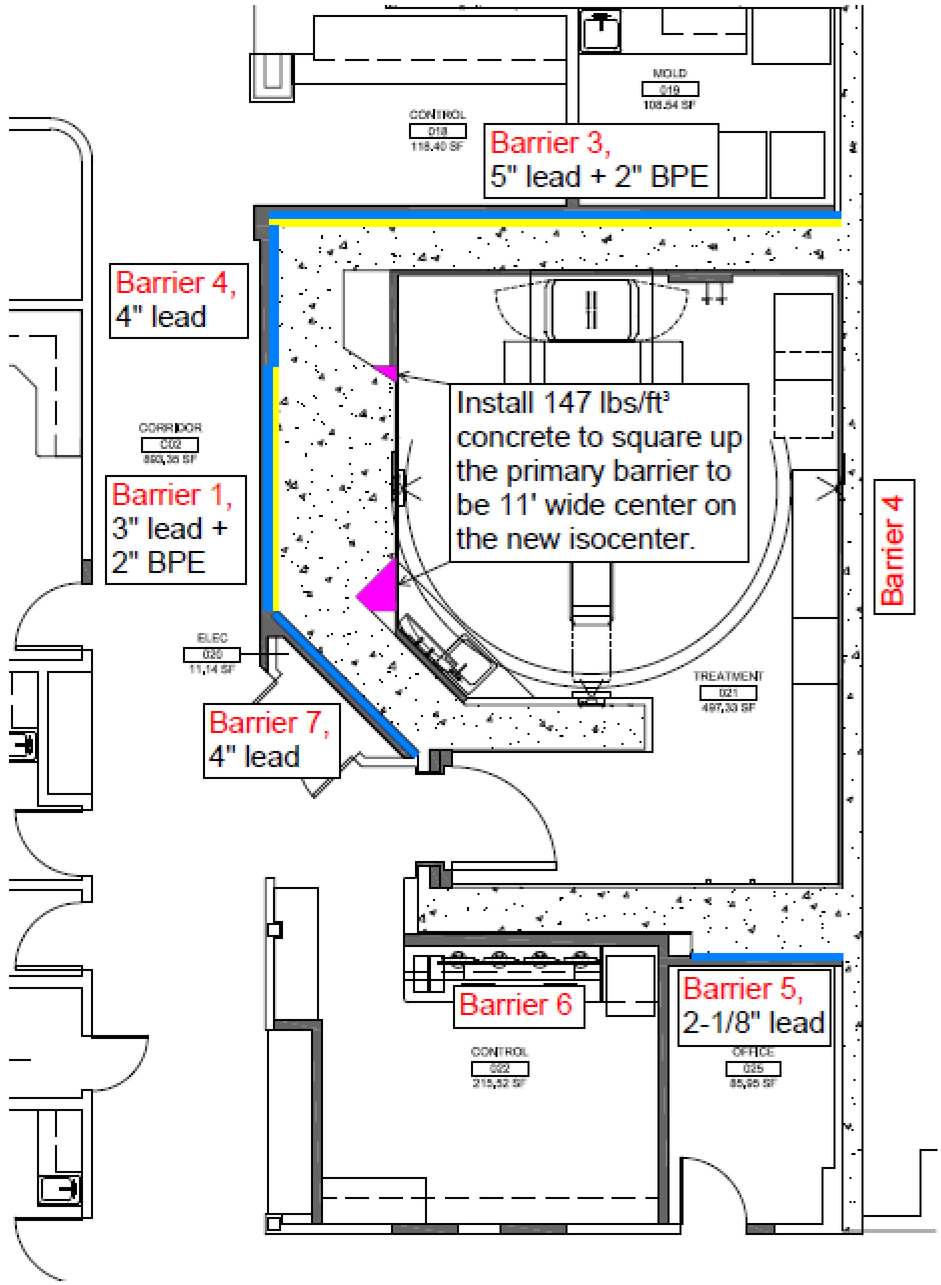
Notes:
1 See attached diagram
2 P= primary barrier, S=secondary barrier
3 BPE = 5% boron (by weight) polyethylene

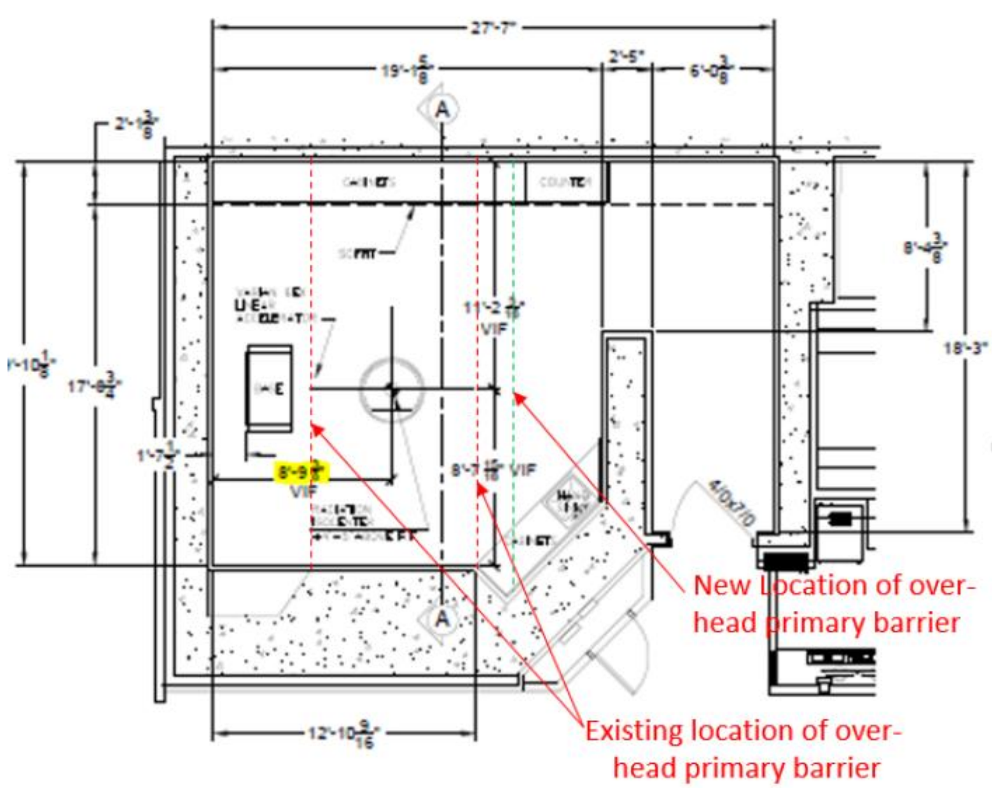
The original floor plan (ie., before the change in Barrier #5)



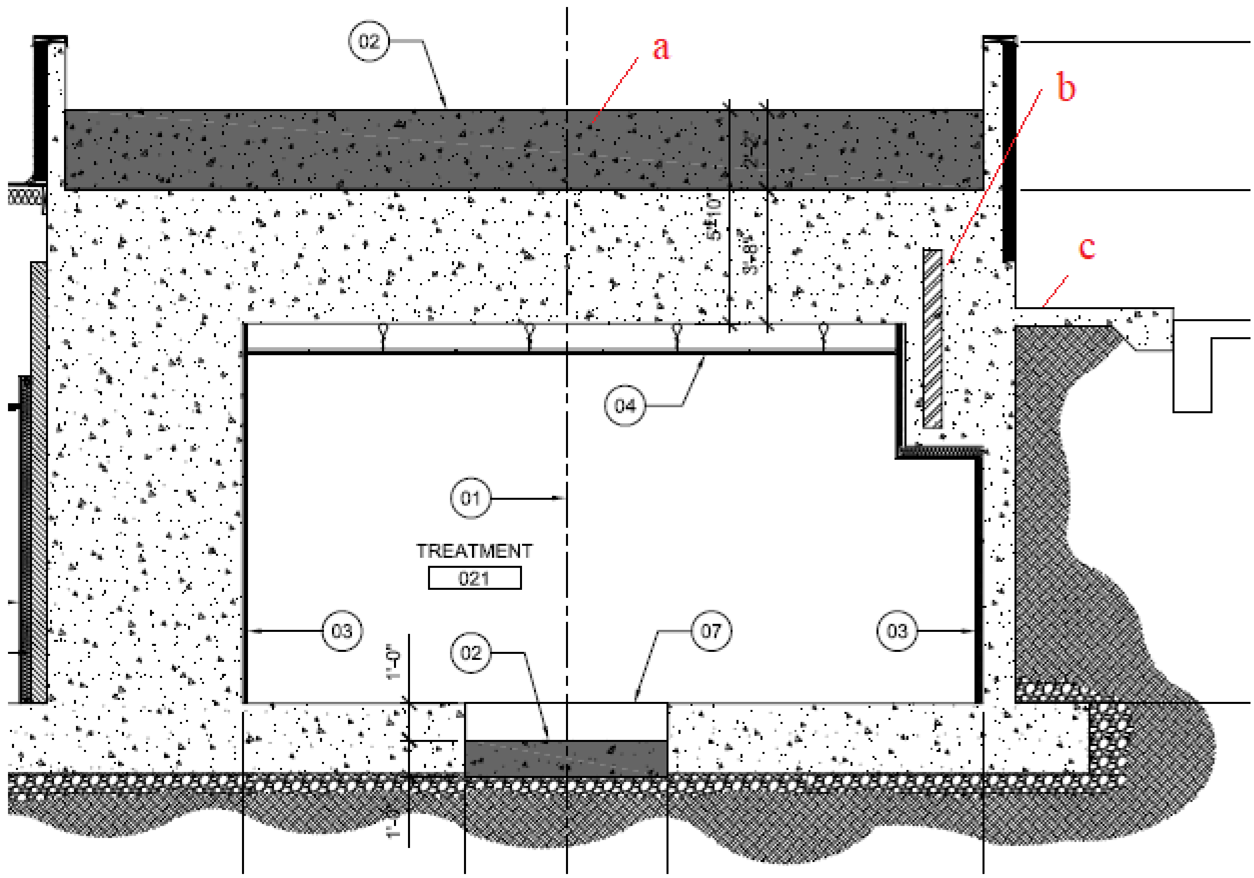
Shielding Floor Plan

The final floor plan (ie., after the change in Barrier #5)





Supplementary Figure 00: Shift in the location of the linac isocenter after the replacement of the linac.



<u>Figure 1:</u> The change in Barrier #8 (roof) and barrier #4 are displayed. The marked "section a" refers to barrier #8 and shows the addition of cast-in-place concrete. Sections b, and c are part of barrier #4 and point to, respectively, an existing embedded steel plate and the structural sidewalk. The centering of the existing embedded steel barrier to the new isocenter is unnecessary. The existing concrete thickness (3'), dimensions of the steel barrier (height x width x depth: 3'-6" x 8'-6" x 8") and the low occupancy factor of that sidewalk leads to adequate rad shielding in that area.

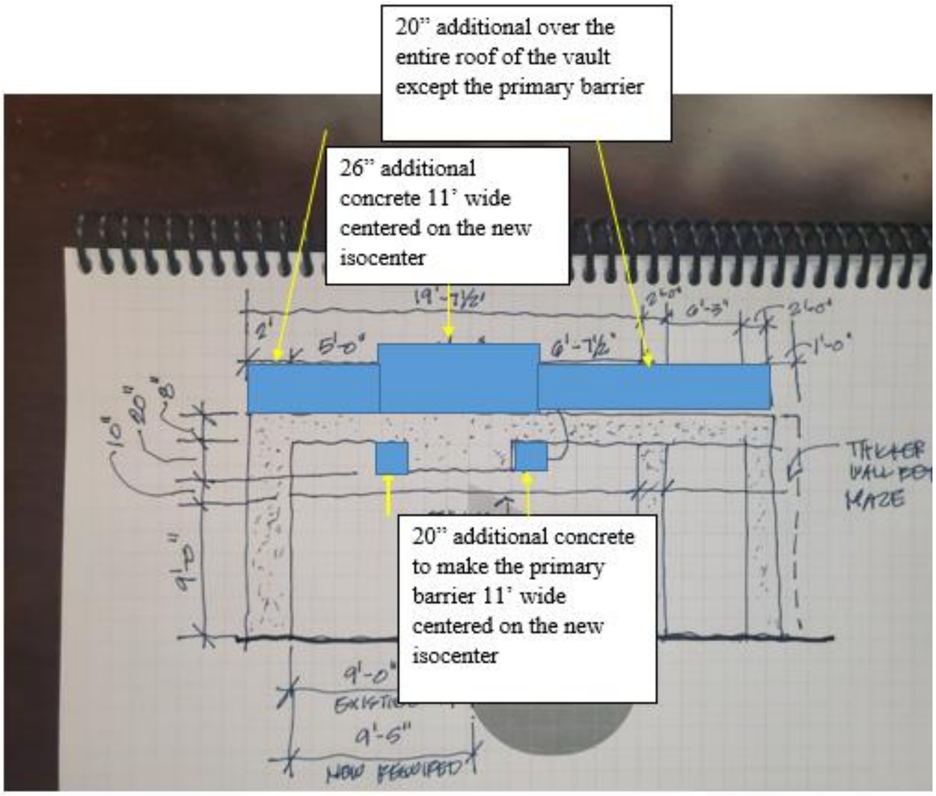
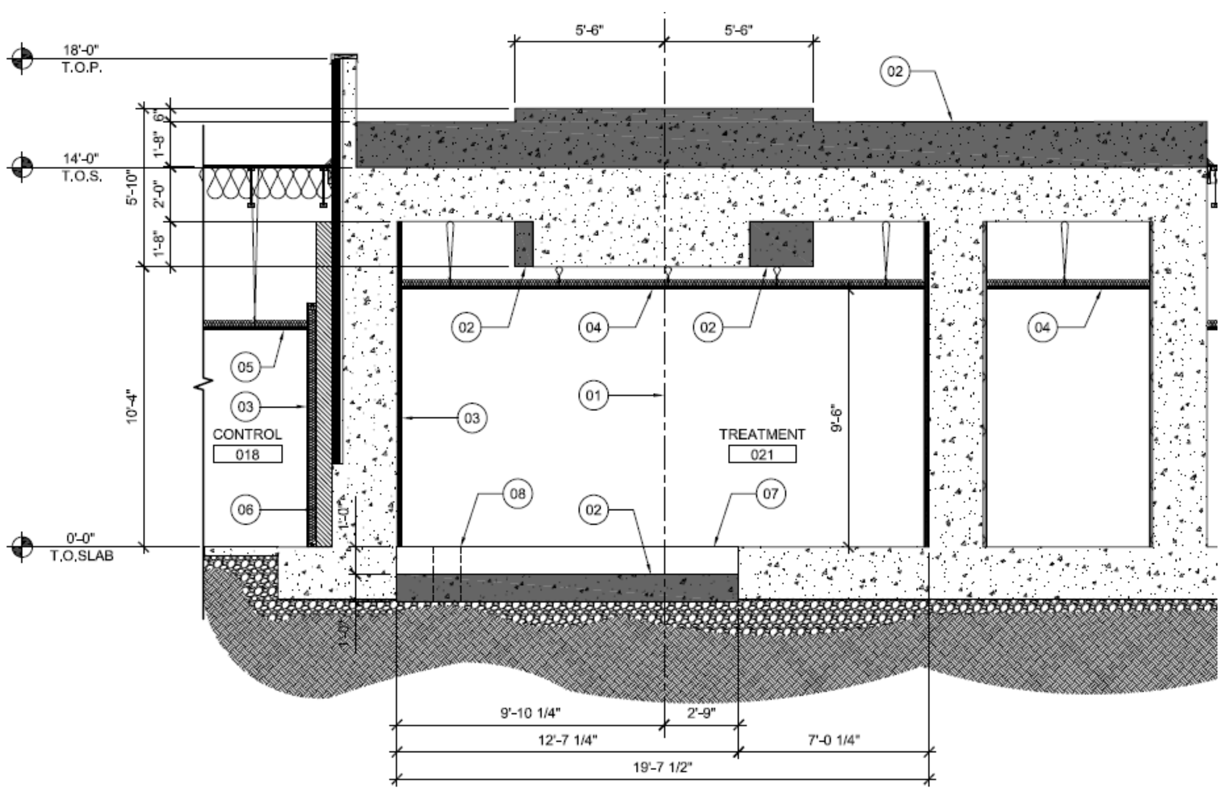


Figure 2: Vertical cross-section view of the vault with distances from the isocenter and updated changes to the additional barriers. The original drawing is at the top. The new architectural drawing is at the bottom. The barrier thicknesses in both figures should now show the same values.



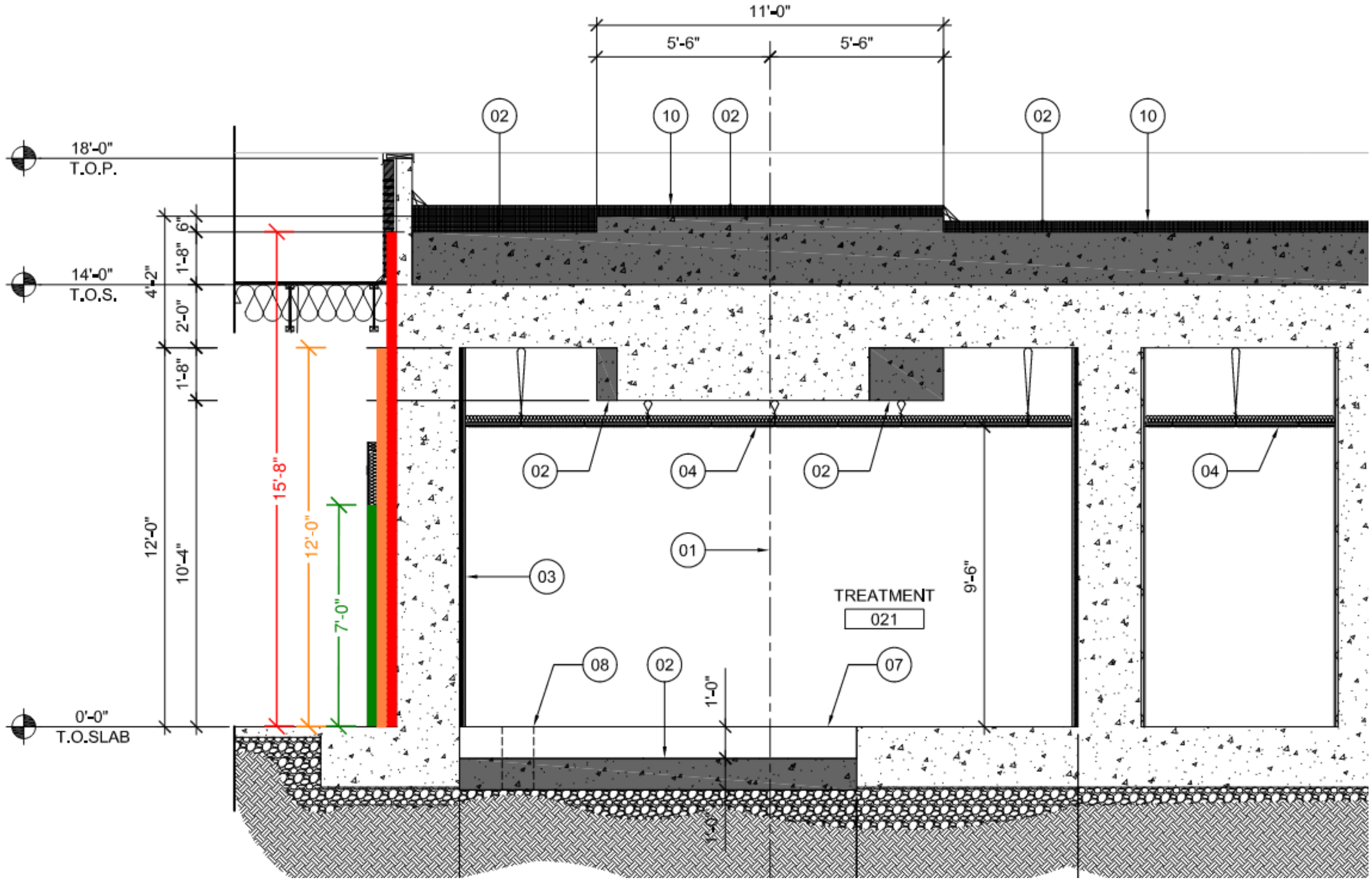


Figure 2b: In ref to revision 5 of this report, the 12’ shield pf Pb+BPE (orange) is recommended around the vault.

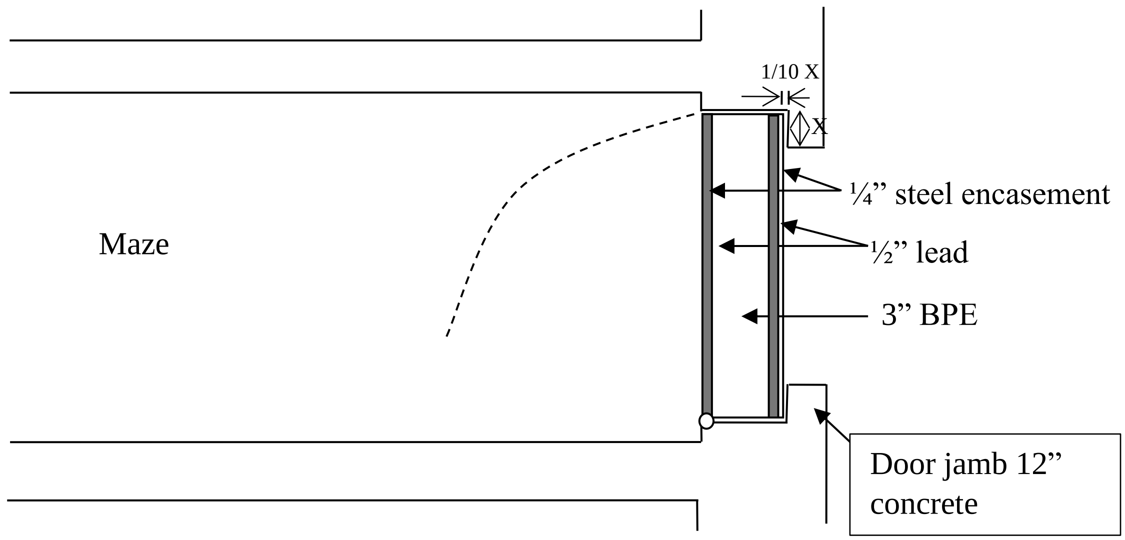
Door Shielding Specifications

Door Recommendation:

Borated polyethylene (BPE) (5% by weight): 3 inch total thickness
Lead: 1 inch total thickness
Steel encasement: ¼ inch
Door construction: ¼” steel + ½” lead + 3” BPE + ½” lead + ¼” steel

Adjacent Wall Recommendation:

1. At least 12” inches of concrete is required for the outside face of doorjamb.
2. The overlap at the sides and top of the door must be at least 10 times the gap between the door and wall.



Above drawing is not to scale.

HVAC

The HVAC ducts must enter above the door in the maze. The ducts must penetrate the barrier at a 45° vertical angle.

Primary Barrier Width Calculation

The required width, w, of the primary barrier was calculated by:

$$d_N = \sqrt{h^2 + (d_{NH} - 1)^2} + 1$$
$$w = \left(\frac{f}{100cm}\right)\sqrt{2}d_N + 0.6m$$

where f is the effective field size, h is the distance from the isocenter to the top of the barrier, and d_{NH} is the distance from the target to the narrow point of the barrier measured horizontally. The barrier width is based on the maximum field size for a Varian True Beam due to the clipping of the corners of the field (35 cm x 35 cm)

Methods of Shielding Calculations

I. Primary Barriers

The primary barrier transmission was calculated with the following equation:

$$B_X = \frac{Pd_{pri}^2}{WUT}$$

where B_X is the barrier transmission factor,
P is the dose per week required outside the barrier,
l_{pri} is the distance om the X-ray target to the point protected,
W is the workload,
U is the use factor,
T is the occupancy factor.

The required number of tenth value layers (TVLs) was calculated with the following equation:

$$n = Log\left(1 / B_X \right)$$

where n is the required number of TVLs

The require barrier thickness, X, was calculated with the following equati

$$X = (n)(TVL)$$

where TVL is the tenth value layer for the barrier shielding material.

The dose rate in any one hour outside the barrier was calculated with the f

$$D_P = \frac{D_{iso} B_X}{d_{pri}^2}$$

where D_P is the dose rate at point protected,
D_{iso} is the dose rate at the isocenter.

II. Secondary Barriers

In order to determine the secondary barrier shielding requirements, head leakage and scatter radiation must be considered. The barrier transmission factor for head leakage was calculated with the following equation:

$$B_l = \frac{\left(\frac{1}{L}\right) P d_{\sec}^2}{WT}$$

where B_l is the transmission factor for head leakage,

L is the photon head leakage,
d_{sec} is the distance from the isocenter to the point protected.

The barrier transmission factor for scatter was calculated with the following equation:

$$B_P = \frac{P\left(\frac{400}{F}\right) d_{sca}^2 d_{\sec}^2}{aWT}$$

where B_P is the transmission factor for radiation scattered from the patient,
d_{sca} is the distance from the x-ray target to the patient,
d_{sec} is the distance from the scattering surface to the point protected,
F is the beam size at patient,
a is the scattering ratio.

The required number of tenth value layers (TVLs) was calculated with the following equation:

$$n = Log\left(1/B_X \right)$$

where n is the required number of TVLs

The require barrier thickness, X, was calculated with the following equation:

$$X = (n)(TVL)$$

where TVL is the tenth value layer for the barrier shielding material.

The barrier thickness for head leakage and patient scatter are compared. If the difference between head leakage and patient scatter is less than a TVL, then a half value layer (HVL) is added to the thicker barrier thickness.

If the radiation is obliquely incident on the shielding barrier, the barrier thickness can be reduced. The radiation path length, t, was calculated by:

$$t = \frac{S}{\cos\theta}$$

where S is the barrier thickness,
θ is the angle of the beam with respect to the normal to the barrier.

III. Width of Primary Barrier

The required width, w, of the primary barrier was calculated by:

$$d_N = \sqrt{h^2 + (d_{NH} - 1)^2} + 1$$

$$w = \left(\frac{f}{100cm}\right)\sqrt{2}d_N + 0.6m$$

where f is the effective field size, h is the distance from the isocenter to the top of the barrier, and d_{NH} is the distance from the target to the narrow point of the barrier measured horizontally. The barrier width is based on the maximum field size for a Varian True Beam due to the clipping of the corners of the field (35 cm x 35 cm)

IV. Maze Door

The total neutron fluence per unit dose of x-ray at the isocenter was calculated with the following equation:

$$\Phi_{total} = \frac{aQ}{4\pi d_1^2} + 6.66\frac{aQ}{2\pi S}$$

where Φ_{total} is the total neutron fluence per unit dose of x-ray at the isocenter,
a is the transmission factor for neutrons that penetrate the head (a=1 for lead, a=0.85 for tungsten),
Q is the neutron source strength,
S is the surface area of the treatment room,
d_1 is the distance from the isocenter to the maze centerline.

The gamma dose, D, at the room door per dose at the isocenter was calculated with the following equation:

$$D = 6.9 \times 10^{-10}(\Phi_{total})10^{-\frac{d_2}{TVD2}}$$

where TVD2 is the tenth value distance,
d_2 is the centerline distance of the maze.

Kersey's method was used to determine the neutron dose equivalent, H, at the outer entrance.

$$H = WH_o\left(\frac{1.41}{d_1}\right)^2 10^{-\frac{d_2}{5}}$$

where W is the workload,
H_o is the neutron dose equivalent at 1.41 meters,
d_1 is the distance from the isocenter to the point on the maze centerline from which the isocenter is just visible,
d_2 is the maze centerline distance.

The required number of tenth value layers (TVLs) for neutrons was calculated with the following equation:

$$n = -\log\left(\frac{\frac{P}{T}}{H}\right)$$

where n is the required number of TVLs

The required barrier thickness, X, was calculated with the following equation:

$$X = (n)(TVL)$$

where TVL is the tenth value layer for polyethylene.

The required number of tenth value layers (TVLs) for neutrons was calculated with the following equation:

$$n = -\log\left(\frac{\frac{P}{T}}{D}\right)$$

where n is the required number of TVLs

The require barrier thickness, X, was calculated with the following equation:

$$X = (n)(TVL)$$

where TVL is the tenth value layer for lead.

*** History of Revisions:**

1. The first radiation shielding report was released on February 12, 2023. Following a TEAMS meeting on June 4, 2014, the questions raised by the architecture team were answered. The changes in the radiation shielding are described in the captions of Figure 1a and Figure 1b.
2. The 2nd radiation shielding report was release on June 4, 2024.
3. The 3rd radiation shielding report was released on July 18, 2024. New information was provided about a few barriers via email and a TEAMS meeting. Accordingly, the values in the Table 1 and those in Figures 1 and 2 were updated. Supplementary Figures 0, and 00 were added to illustrate, respectively, the orientation of the cancer center and the shift in the location of the isocenter after the planned linac replacement.
4. Following release of report v 3b, customer converted the space behind barrier # 5 (“dark room”) to an office space with occupancy factor of 1.
5. Responded to architect’s inquiry email (Nov 6 and 12). In view of the updated distances, the 12’-high shield is recommended (see figure 2b)